PHY102 Electricity and Magnetism Jan-April 2026: Assignment 4

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To be discussed in the Tutorial on Feb 20

1. Consider the vector fields:

$$\vec{V_1}(x, y, z) = x\hat{x} + y\hat{y}$$
$$\vec{V_2}(x, y, z) = x\hat{x} - y\hat{y}$$
$$\vec{V_3}(x, y, z) = y\hat{x} + x\hat{y}$$
$$\vec{V_4}(x, y, z) = y\hat{x} - x\hat{y}$$

   (a) Calculate the curl and divergence of these vector fields, plot the fields lines to develop an intuition.

   (b) For the last case how do you visualise uniform nonzero curl while the field looks quite different in different regions?

   (c) Write down a vector field (in $x - y$ plane for simplicity) where curl changes sign. Plot the vector arrow of this field.

2. Consider Laplace equation in 1 dim. It reads

$$\frac{d^2 f(x)}{dx^2} = 0$$

. Imagine boundaries at $x = a$ and $x = b$. Argue that the solution cannot have a maxima or minima inside the domain $a \leq x \leq b$. If the values $f(a)$ and $f(b)$ are fixed, do we have a unique solution?

3. Now consider Laplace equation in 2 dim. This will read

$$\frac{\partial^2 f(x, y)}{\partial x^2} + \frac{\partial^2 f(x, y)}{\partial y^2} = 0.$$

   (a) Convert this equation into plane polar coordinates (r, θ), with $x = r\cos\theta$ and $y = r\sin\theta$.

   (b) Solve this equation in the region $r_1 \leq r \leq r_2$ with boundary conditions $f(r_1, \theta) = \phi_1$ and $f(r_2, \theta) = \phi_2$.

4. Consider the problem of stabilising positive charge q_0 at the origin by placing charges around it. To begin with, place four charges q on a square in the $x - y$ plane such that q_0 is at the center of the square. Is this configuation stable in he $x - y$ plane? How can it be made stable? Imagine a ring of uniform charge in the $x - y$ plane, the charge q_0 at the center. Is this configuation stable in the $x - y$ plane? Is it stable along z? Can you try to make it stable in three dimensions?

5. Consider a vector field $\vec{A}(x, y, z)$ Prove that $\nabla.(\nabla \times A) = 0$ (Divergence of a Curl is always zero.) Prove using partial derivates in cartesian coordinates and also try to prove by a coordinate independent way. (Hint: See Purcell book problem 2.16 and the hint there.)

6. A flat nonconducting sheet lies in the $x - y$ plane. The only charges in the system are on this sheet. In the half-space above the sheet, $z > 0$ the potential is $\phi = \phi_0 e^{-kz} \cos kx$ and where ϕ_0 and k are constants.

   (a) Verify that ϕ satisfies Laplace's equation in the space above the sheet.

   (b) What do the electric field lines look like?

   (c) Describe the charge distribution on the sheet.